

Commuting ribbon operators: a classification of the weights

[SKELETON — 17 September 2026. Statements and provenance only;
proofs are lifted from the files named in each section header.]

Clio

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Abstract

For $1 \leq e < f$ let R_e^W be the operator on symmetric functions that adds a connected e -ribbon with a weight W depending on the whole shape of the ribbon. We determine every pair $(W, \bar{W})$ with $[R_e^W, R_f^{\bar{W}}] = 0$. Writing $d = \gcd(e, f)$, the answer is a *tensor* statement: the commuting pairs are exactly those obtained from a single rank- d *seed* weight by concatenation, subject to one quadratic condition. Two consequences: a dichotomy — a nonzero commuting weight either vanishes nowhere or vanishes on at least three quarters of the ribbon shapes of its rank, with nothing in between, and both extremes attained — and, for every $d \geq 3$, operators that add *one ribbon shape and nothing else* and still commute. For $d \leq 2$ no such operator exists and the classical Murnaghan–Nakayama sign is forced. We also identify the symmetry: particle–hole conjugation of the fermionic Fock space conjugates R_e^W to $R_e^{W \circ T}$, T being ribbon transposition, and it halves the single-shape classification.

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Status of this file. This is the skeleton demanded by `state/PROVE.md` of 17 September 2026, Part B. It fixes the paper’s notation once (§2), states the results in full (§3), and records for each later section which artifact the proof is lifted from, by filename and label. Nothing here is proved; everything here is *stated* in the paper’s notation, which is not the notation of any one source. §6 is the honest list of what remains open, and §5 lists the four places where a source statement must be changed before it is lifted.

1 Introduction

To be written last. The arc: Murnaghan–Nakayama says that multiplying by a power sum adds ribbons with the sign $(-1)^{\text{ht}}$, and that these operators commute. Ask which *other* weights make them commute. If the weight may see only the height of the ribbon, the answer is: none but Murnaghan–Nakayama (§4). If it may see the whole shape, the answer is strictly larger as soon as $\gcd(e, f) \geq 3$, and the extra solutions are not deformations of Murnaghan–Nakayama — they are operators with large zero sets, the extreme case adding a single ribbon shape (§4). The organising fact is that the one-bead sector of the commutator is literally the statement that two words commute in a free monoid, so the gcd in the answer is the gcd of Lyndon–Schützenberger.

2 Notation, fixed once

Convention 2.1 (the objects). K is an integral domain with fraction field F . $\Lambda = \Lambda_K$ is the ring of symmetric functions with Schur basis $\{s_\lambda\}$ and Hall inner product. For a partition λ , $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$ is its *Maya set*; elements are *beads*, elements of $\mathbb{Z} \setminus M$ *holes*. $\mathcal{F}$ is the charge-0 semi-infinite wedge with basis $|M\rangle$, identified with Λ by $|M(\lambda)\rangle \leftrightarrow s_\lambda$; the free fermions are undeformed, $\{\psi_a, \psi_b^*\} = \delta_{ab}$, and $n_a = \psi_a \psi_a^*$.

Convention 2.2 (words, shapes, weights). A legal *rank- e move* in M is $b \mapsto b + e$ with $b \in M$, $b + e \notin M$; its *occupancy word* is $u \in \{0, 1\}^{e-1}$, $u_i = [b + i \in M]$. We write $|u| = \sum u_i$ and $\text{ht} = |u|$, the number of rows of the ribbon minus one. A *shape weight of rank e* is a function $W : \{0, 1\}^{e-1} \rightarrow K$; it defines

$$R_e^W s_\lambda = \sum_{\mu} W(u) s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected e -ribbon of occupancy word u . The *sign-normalised* weight is $\sigma(u) := (-1)^{|u|} W(u)$; Murnaghan–Nakayama is $\sigma \equiv \text{const}$, i.e. $R_e^W = \sigma_0(p_e \cdot)$. A *height weight* is one with $W(u)$ depending only on $|u|$.

Convention 2.3 (the three involutions of $\{0, 1\}^n$). $(\text{rev } u)_i = u_{n+1-i}$, $(\text{cpl } u)_i = 1 - u_i$, and $T := \text{cpl rev} = \text{rev cpl}$, so $(u^T)_i = 1 - u_{n+1-i}$. T is ribbon transposition. $W^\vee := W \circ T$.

Convention 2.4 (concatenation). For shape weights σ_1, σ_2 of ranks m_1, m_2 ,

$$(\sigma_1 \otimes \sigma_2)(y) := \sigma_1(y_{[1, m_1-1]}) \sigma_2(y_{[m_1+1, m_1+m_2-1]}), \quad y \in \{0, 1\}^{m_1+m_2-1},$$

the letter y_{m_1} being ignored; $\otimes$ is associative, and $\sigma^{\otimes k}$ is the k -fold power.

Convention 2.5 (the two quadratic conditions). A rank- m shape weight σ satisfies **(2B $_m^*$)** if for all $\delta \in [1, m-1]$, $\pi, \varsigma \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{m-\delta-1}$,

$$\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) = \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma);$$

and **(2B $_m^{**}$)** if the same holds with the second factor's prefix decoupled. **(2B **)** $\Rightarrow$ **(2B *)**, strictly from $m \geq 4$.

Convention 2.6 (border-split words). Let $w \in \{0, 1\}^n$. An integer $p \in [0, n - 1]$ is a *border length* of w if $w_{[1,p]} = w_{[n-p+1,n]}$ (so $p = 0$ always is one). Call w **border-split** if

$$w_{p+1} \neq w_{n-p} \quad \text{for every border length } p \text{ of } w. \quad (\text{BS})$$

Remark 2.7 (why not “non-overlapping”). [6] calls (BS) “non-overlapping”. That term is already taken: in combinatorics on words a non-overlapping (= unbordered, bifix-free) word is one with no border length $p \in [1, n - 1]$ at all. (BS) is strictly weaker — it permits a border provided the two continuation letters differ. The families differ from $n = 4$ ($d = 5$) on: 8 border-split words against 6 unbordered ones, the difference being 0101 and 1010. The counts 0, 2, 4, 8, 12, 28, 48, 104 for $d = 2, \dots, 9$ reported in [6, §9] are the (BS) counts, and are correct; only the name is hazardous. **This paper imports the condition and not the word.**

2.1 Reconciliation with the five sources

The five artifacts do not use one convention. The table records every symbol whose meaning changes between them. A row marked $\blacklozenge$ is a genuine collision — the same symbol naming two different objects — and is the reason this section exists.

symbol	Q91 / Q92 / Q147	Q149	Q150c1 / Q150c2	this paper
$\blacklozenge \setminus \mathbb{F}$	$\mathcal{F}$, the Fock space	—	$\mathbb{F}$, a <i>field</i>	$\setminus \text{Fock} = \mathcal{F}$ (space), $\setminus \text{Fld} = F$ (field)
$\blacklozenge \mathcal{C}$	—	—	$\mathcal{C}_{j,b}$, cylinder sets ($\setminus \text{Cset}$)	$\mathcal{C}_{j,b}$ cylinders; particle-hole conjugation is $\mathcal{P}$, <i>never</i> $\mathcal{C}$
$\blacklozenge \tau$	Q147: τ_N , the height weight in fermionic coordinates, $W_N = (-1)^N \tau_N$	—	the rank- d <i>seed</i> weight of the tensor decomposition	τ = seed only. Q147’s τ_N is eliminated: we state Q147 in W/σ coordinates throughout
$\blacklozenge w$	Q147: w_N , a height weight	—	Q150c2: $w \in \{0, 1\}^n$, a <i>word</i>	w = word only; height weights are W with $W(u) = W_{ u }$
$\blacklozenge \gamma$	—	$\gamma : \mathbb{Z}/d \rightarrow F^\times$, a character	Q150c1/c2 proofs: $\gamma \in \{0, 1\}$, a single <i>letter</i>	γ = character; letters are b
$\blacklozenge \alpha$	—	$\alpha(u)$, the composition (§1); <i>and</i> $\alpha \in K^\times$, a scalar (§5)	$\alpha(w)$, the composition	$\alpha(u)$ = composition; scalars are $c, \bar{c}$
$\blacklozenge k$	Q92: $k_{b,c}^{e,f}$, the interference index	—	Q150c2: k , the tensor power	k = tensor power; the interference index is $\varkappa_{b,c}^{e,f}$
$\blacklozenge m$	$m(x) = [x \in M]$; Q147 $m_e = m(b+e)$	—	m_1, m_2 , <i>ranks</i>	$m(x)$ = bead indicator; ranks are e, f, d

symbol	Q91 / Q92 / Q147	Q149	Q150c1 / Q150c2	this paper
$\diamond N$	$N_{(x,y)} = \sum_{x < j < y} n_j$, an operator; Q147 N = a height index	—	—	$N_{(x,y)}$ operator only; height indices are h
$\backslash \text{Mult}\{\mathbf{p_e}\}$	M_{p_e}	$p_e \cdot$	$p_e \cdot$	$p_e \cdot$ (macro dropped)
height macro	$\backslash \text{hgt}$	$\backslash \text{hgtt}$	$\backslash \text{height}$ (c1), $\backslash \text{hgt}$ (c2)	$\backslash \text{hgt}$, one macro
the operator	$R_e(t)$ (weight t^{ht} built in)	R_e^W	R_e^W	R_e^W ; $R_e(t) := R_e^W$ with $W(u) = t^{ u }$
σ	—	$\sigma(u) (-1)^{ u } W(u)$	= same; $\sigma, \bar{\sigma}$ of ranks e, f	same
d	—	$d = \gcd(e, f)$	= $\gcd(e, f)$ (c1); the rank of a single weight (c2)	$d = \gcd(e, f)$ always; a bare rank is m
n	—	—	Q150c2: $n = d - 1$, the word length	n = word length, stated each time
(NB)	—	—	Q150c2 redefines “non-overlapping”	(BS), border-split (Rem. 2.7)

Remark 2.8 (one sign convention, stated once). Q147 records $W_N = (-1)^N \tau_N$: the $(-1)^N$ inside $(-t)^N$ in the fermionic normal form is not part of the weight, it cancels the Jordan–Wigner sign. We never write τ_N , so this identity is folded into Convention 2.2 and never reappears. The one place the sign is visible is $\sigma = (-1)^{|u|} W$: in σ -coordinates every matrix element of the commutator is a difference of two products with $+$ signs.

3 The results

Theorem 3.1 (classification). *Let $1 \leq e < f$, $d = \gcd(e, f)$, and let $W, \bar{W}$ be shape weights of ranks e, f over K with $W \not\equiv 0 \not\equiv \bar{W}$. Then $[R_e^W, R_f^{\bar{W}}] = 0$ if and only if there are a nonzero shape weight τ of rank d over F and $c, \bar{c} \in F^\times$ with*

$$\sigma = c \tau^{\otimes(e/d)}, \quad \bar{\sigma} = \bar{c} \tau^{\otimes(f/d)},$$

and σ satisfies (2B $_e^$) (vacuous for $e = 1$).*

Theorem 3.2 (the dichotomy). *Let $\tau \not\equiv 0$ be a rank- d shape weight over K satisfying (2B $_d^*$). Then either τ vanishes nowhere, or*

$$\# \text{supp}(\tau) \leq 2^{d-3} = \frac{1}{4} \cdot 2^{d-1},$$

i.e. τ vanishes on at least three quarters of the 2^{d-1} ribbon shapes of rank d . There is nothing in between. Both alternatives are attained, and the bound 2^{d-3} is attained, by the indicator of a cylinder $\mathcal{C}_{j,b} = \{w : w_j = b, w_{d-j} = 1 - b\}$. For $d \leq 2$ only the first alternative occurs.

Theorem 3.3 (single-shape commuting operators). *Let $d \geq 3$, let $w \in \{0, 1\}^{d-1}$ be border-split, let $\alpha(w) \models d$ be the corresponding ribbon shape, let $k \geq 2$ and put $e = d$, $f = kd$. Let $W = \mathbb{K}_w$ and let $\bar{W}(y) = (-1)^{\text{ht}(y)} \prod_{i=1}^k [y_{[(i-1)d+1, id-1]} = w]$. Then R_d^W adds the single ribbon shape $\alpha(w)$ and nothing else, $\bar{W}$ is supported on exactly 2^{k-1} of the 2^{kd-1} shapes of rank kd , and $[R_d^W, R_{kd}^{\bar{W}}] = 0$. The number of admissible w is 2, 4, 8, 12, 28, 48, 104 for $d = 3, \dots, 9$.*

Theorem 3.4 (the symmetry). *Let $\mathcal{P}(M) = \{-1 - x : x \in \mathbb{Z} \setminus M\}$ be particle-hole conjugation, an involution of charge-0 Maya sets realising $s_\lambda \mapsto s_{\lambda'}$. Then for every e and W ,*

$$\mathcal{P} R_e^W \mathcal{P} = R_e^{W \circ T},$$

*so $X \mapsto \mathcal{P}X\mathcal{P}$ is an algebra automorphism preserving the commuting locus. On the nowhere-zero part of Theorem 3.1 it acts by a scalar; on the single-shape part it acts by the involution $w \mapsto w^T$ of border-split words, whose fixed points are exactly the antipalindromes, i.e. exactly the w for which $\mathbb{K}_w$ satisfies $(2B_d^{**})$. Hence the single-shape classification may be carried out on T -orbits: 1, 2, 2, 6, 6, 18, 24, 60 orbits for $d = 2, \dots, 9$.*

Corollary 3.5 (the coprime headline). *If $\gcd(e, f) = 1$ then τ has rank 1, so σ and $\bar{\sigma}$ are constant: $R_e^W = c(p_e \cdot)$ and $R_f^{\bar{W}} = \bar{c}(p_f \cdot)$. Shape dependence is forced away, and with it every parameter but one per rank.*

Corollary 3.6 (height weights). *Restricted to height weights the classification collapses for all $e < f$: $[R_e^W, R_f^{\bar{W}}] = 0$ with both nonzero forces $W(u) = (-1)^{|u|}W_0$ and $\bar{W}(y) = (-1)^{|y|}\bar{W}_0$.*

4 Section skeleton and provenance

Each subsection below states what the corresponding section of the paper proves and the file and label the proof is lifted from. Numbers are given from the label map of Appendix A, generated from the .aux files on 17 September 2026.

§4 The abacus dictionary and the fermionic normal form

Proves: adding an e -ribbon is a bead move $b \mapsto b + e$; the height is the number of beads jumped; the occupancy word determines the ribbon shape, *top row first*, by $\alpha(u) = (c_{k+1} - c_k, \dots, c_1 - c_0)$ where the c_i list $\{0\} \cup \{i : u_i = 1\} \cup \{e\}$; and $R_e(t) = \sum_b \psi_{b+e} \psi_b^* (-t)^{N(b, b+e)}$. *Lifted from:* 2026-09-06-c2-Q91-fermionic-normal-form.tex, lem:abacus, lem:sign, lem:dressing, thm:main (=Thm. 7); and 2026-09-15-c1-Q149-shape-weights.tex, lem:dict (=Lem. 1.3). *Note:* the reversal in $\alpha(u)$ is not cosmetic — without it the two L-trominoes exchange names, which is exactly the distinction the cheapest test of the paper turns on.

§5 The two sectors of the commutator

Proves: $\langle s_\mu, [R_e^W, R_f^{\bar{W}}] s_\lambda \rangle = 0$ unless $|M(\lambda) \triangle M(\mu)| \in \{2, 4\}$; closed forms for the one-bead and two-bead matrix elements in σ -coordinates. *Lifted from:* 2026-09-15-c1-Q149-shape-weights.tex, lem:sectors (=Lem. 4.1), lem:onebead (=Lem. 4.2), lem:twobead (=Lem. 4.3). *Background, for the single- t case only:* 2026-09-07-Q92-cross-rank-commutator.tex, thm:matrix (=Thm. 5), giving $(m(b+e) - m(b+f))(1+t)t^{N-1}$ in the one-bead sector and $t^{P+Q}(t^{-z} - t^z)$ in the two-bead sector, and thm:main (=Thm. 6), $[R_e(t), R_f(t)] = -\frac{1+t}{t}(\Phi_{e,f} + (t-1)\Psi_{e,f})$. *Note:* the witness $\langle s_\mu, [R_e(t), R_f(t)] s_\lambda \rangle = t^f - t^{f-2}$ is prop:psiwitness (=Prop. 16) and belongs to the **two**-bead sector; the brief for this session filed it under the one-bead sector.

§6 Height weights force Murnaghan–Nakayama

Proves: Corollary 3.6, using only the one-bead sector and only the occupancy pattern $m_e = 1$, $m_f = 0$. *Lifted from:* 2026-09-11-c2-Q147-multi-t-bracket.tex, thm:main (=Thm. 7) via lem:B, with lem:realise as the control that every window configuration occurs. *Note:* restate in

W -coordinates; do not import τ_N (Remark 2.8). The $(1+t)$ factor of Q92 is the $N = 0$ instance of the forcing relation $W_{N+1} = -W_N$.

§7 The one-bead sector is commutation in a free monoid

Proves: the one-bead sector vanishes identically iff $\bar{\sigma} = \sigma \otimes \rho = \rho \otimes \sigma$ for a unique nonzero ρ of rank $f - e$; and two shape weights commute under $\otimes$ iff both are powers of a common seed — Lyndon–Schützenberger for the semigroup of Convention 2.4, by Euclidean descent. This is where the gcd comes from. *Lifted from:* 2026-09-16-c1-Q150-nonvanishing.tex, prop:1B (=Prop. 3.1) and thm:structure0 (=Thm. 1.2).

§8 The two-bead sector is $(2B^*)$, and the classification

Proves: given the one-bead conclusion, the two-bead sector is exactly $(2B_e^*)$ for σ ; hence Theorem 3.1. *Lifted from:* 2026-09-16-c1-Q150-nonvanishing.tex, prop:2B (=Prop. 3.3) and thm:main0 (=Thm. 1.3). *Note:* prop:2B’s converse direction makes a *second, separate* appeal to K being a domain, with witness $\rho(\psi_0)$; say so, per [8, one-domain-appeal].

§9 The structure of an admissible support, and the dichotomy

Proves: an admissible support is empty, everything, or contained in a cylinder $\mathcal{C}_{j,b}$ with $2j \neq d$; the maximal proper admissible sets are exactly the $2\lfloor(d-1)/2\rfloor$ cylinders; hence Theorem 3.2. Also: $d \geq 3$ is exactly the condition that $[1, d-1]$ contains some $j \neq d-j$, which is the conceptual content of the threshold $\gcd(e, f) \geq 3$. *Lifted from:* 2026-09-16-c2-Q150-H2-single-shape.tex, def:S (=Def. 2.2), prop:supp (=Prop. 2.3), thm:struct (=Thm. 6.1), cor:three (=Cor. 6.2), cor:maximal (=Cor. 6.3), cor:quarter (=Cor. 6.4). *Correction required:* see §5(C3).

§10 Single-shape weights

Proves: $\mathcal{K}_w$ satisfies $(2B_d^*)$ iff w is border-split; it satisfies $(2B_d^{**})$ iff w is an antipalindrome, of which there are $2^{(d-1)/2}$ for d odd and none for d even; hence Theorem 3.3, with the smallest new case $d = 4, k = 2, w = 001$, shape $(1, 3)$. *Lifted from:* 2026-09-16-c2-Q150-H2-single-shape.tex, def:border (=Def. 4.1), thm:singleton (=Thm. 4.2), rem:endrows (=Rem. 4.3), thm:singleton2 (=Thm. 4.5), thm:main (=Thm. 5.1), cor:48 (=Cor. 5.2). *Correction required:* see §5(C1). *Note:* for even d these operators are new precisely because the $(2B^{**})$ route produces none of them — there are no antipalindromes of even length $n = d - 1$.

§11 Generation: the admissible sets form a closure system

Proves: Z is admissible iff closed under the flip rule $u, v \in Z$ δ -linked $\Rightarrow u^{(\delta)}, v^{(n-\delta+1)} \in Z$; so the admissible sets are the closed sets of a Moore family. *Lifted from:* 2026-09-16-c2-Q150-H2-single-shape.tex, def:linked (=Def. 7.1), thm:closure (=Thm. 7.2), prop:moore (=Prop. 7.3).

§12 Particle–hole conjugation

Proves: Theorem 3.4; that $\mathcal{P}$ -conjugation is an *automorphism*, not an antiautomorphism, so it is not Terwilliger’s $\dagger$; and that no invertible graded linear map implements rev or cpl alone, so T is exactly the part of the word-level $(\mathbb{Z}/2)^2$ that an operator can see. *Lifted from:* 2026-09-17-c1-Q151-particle-hole.tex, thm:A, lem:Ctrans, thm:C, thm:D, cor:parity, prop:norev, cor:accounting. *Note:* thm:A also upgrades [4, lem:transpose] (=Lem. 1.6) from computed to proved: the fact that transposition is

reverse-and-complement on occupancy words, previously checked on 1500 ribbons, has a three-line proof on the bead lattice.

5 Corrections to the sources, to be applied before lifting

Four statements must change on their way into the paper. None is a gap in a proof; three are glosses or names that overstate or collide, and one is a superseded theorem that must not be quoted.

- (C1) **Do not import the word “non-overlapping.”** [6, Def. 4.1] redefines it to mean (BS); the standard meaning is strictly stronger and the two differ from $d = 5$ on. Use Convention 2.6. (Remark 2.7.)
- (C2) [4, Thm. 5.2] **is superseded and must not be quoted as the classification.** Its `lem:nonvanish` (=Lem. 8.2), “a nonzero commuting weight vanishes nowhere”, is **false** for $\gcd(e, f) \geq 3$ [5, Cor. 1.4], and [4] itself flags it as “the one gap of the paper” for $e \geq 3$, adding “everything after this point assumes it”. That warning is exact, and the audit below says precisely how far it reaches — the blanket instruction “re-check all of [4, §8]” is too coarse to act on, and two of its five results are unconditional.

label	number	status for the write-up
<code>lem:flip</code> (F0)–(F3)	8.1	Unconditional; lift freely. Its proof needs only $\sigma \neq 0 \neq \bar{\sigma}$ and K a domain — it picks <i>some</i> nonzero witness $\sigma(A)$ or $\sigma(D)$, never all of them.
<code>lem:nonvanish</code>	8.2	False for $\gcd(e, f) \geq 3$. True exactly for $\gcd(e, f) \leq 2$ [5, Thm. 6.1]. Note this <i>extends</i> [4]’s own proved range, which was $e \leq 2$: e.g. $(e, f) = (4, 6)$ has $e > 2$ but $\gcd = 2$.
<code>lem:const</code>	8.3	Conditional — stated “granting Lemma 8.2”. Valid exactly for $\gcd(e, f) \leq 2$.
<code>lem:char</code>	8.4	Conditional , likewise.
<code>prop:e1</code>	8.5	Unconditional; lift freely. Its proof is a cyclic-shift argument on the one-bead relation and never mentions nonvanishing.
<code>prop:suff</code>	7.1	Unconditional and still true: the γ -family does commute. This is the sufficiency half of Thm. 5.2 and survives intact — it is the nowhere-zero part of Theorem 3.1.
Thm. 5.2, necessity	5.2	Valid exactly for $\gcd(e, f) \leq 2$. Superseded in general by [5, Thm. 1.3], which reaches the same conclusion for $\gcd \leq 2$ and the correct larger one for $\gcd \geq 3$, <i>using no nonvanishing input at all</i> .
<code>cor:size</code>	5.4	True as stated about the group of admissible γ , but its <i>title</i> (“the size of the locus”) is wrong for $d \geq 3$: it sizes the γ -torus, which is then only the nowhere-zero part.

So the paper lifts 8.1, 8.5 and 7.1 from [4] and takes everything else from [5].

- (C3) **“More than three quarters” must become “at least three quarters.”** The display in [6, Cor. 6.4] is $\#\text{supp}(\tau) \leq 2^{d-3}$, which is correct and *attained*: the indicator of a cylinder $C_{j,b}$ has support exactly 2^{d-3} and satisfies $(2B_d^*)$ (checked for all j, b and $3 \leq d \leq 8$). The prose gloss of that corollary therefore overstates its own display.
- (C4) **The $t^f - t^{f-2}$ witness is two-bead, not one-bead.** [2, Prop. 16] derives it from $t^{P+Q}(t^{-\varkappa} - t^\varkappa)$ with $P = 0$, $Q = f - 1$, $\varkappa = -1$. The one-bead element of that paper is $(m(b+e) - m(b+f))(1+t)t^{N-1}$.

6 What is not proved

- (H1) **The exact rank- d form of $(2B_e^*)$ when $e > d$.** Theorem 3.1 imposes $(2B_e^*)$ on $\sigma = c\tau^{\otimes(e/d)}$, which is what the commutator hands over; but which *seeds* τ have $\tau^{\otimes k}$ satisfying $(2B_{kd}^*)$ is not known in closed form. What is known [5, Rem. 5.2]: $(2B_d^{**}) \Rightarrow (2B_e^*) \Rightarrow (2B_d^*)$, and *both* implications are strict for $d \geq 4$, so the middle condition is genuinely new. Whether it depends on k is unknown; it does not for $k \in \{2, 3\}$ at $d \leq 4$. Theorem 3.3 is the first clean fact about it from the other side: for even d *no* single-shape τ satisfies $(2B_d^{**})$ — there are no antipalindromes of odd length — yet single-shape τ at $d = 4, 6, 8$ do give commuting pairs. That is exactly why those examples are new, and it is also why the $(2B^{**})$ route of [5, Thm. 6.6] cannot reach them. Note this does not weaken any theorem above; it is a question about a *sufficient* condition, not about the classification. Theorem 3.4 adds one constraint the answer must satisfy: the middle condition is T -stable, since $(2B^*)$ is and $\otimes$ commutes with T .
- (H2a) **The interior of a cylinder is not described.** Theorem 3.2 bounds an admissible support by a cylinder and [6, Cor. 6.3] shows the cylinders are the maximal proper ones, but the admissible subsets *inside* a cylinder are not classified. The counts are 1, 3, 11, 31, 356. The obvious induction is **closed**: there is no uniform maximal admissible set one level down — at $m = 4$ the maximal proper admissible subsets of a cylinder have sizes $\{4, 4, 5, 5, 5, 5, 6, 6, 7, 9, 13\}$, so no single object plays the role the cylinders play one level up. [6, thm:closure] gives a generation rule but not a normal form.
- (H2b) **No uniform description of the value lattice L_Z .** Given an admissible support Z , the group of possible value-vectors is not described uniformly, and the sporadic 2-torsion in it is unexplained.
- (H2c) **The T -fixed locus is known only at the two extremes.** Theorem 3.4 determines the action of T on the nowhere-zero seeds (a scalar) and on the single-shape seeds (the free involution $w \mapsto w^T$). For a seed with zeros that is not a single shape, whether $\tau \circ T \in F^\times \tau$ is open.
- (H2d) **Reversal alone.** [7, prop:norev] rules out invertible *degree-preserving* implementations of rev; a degree-mixing one is not ruled out. Since rev acts nontrivially on the locus (it inverts γ), an implementation would be worth more than $\mathcal{P}$ is.

A Label $\rightarrow$ number map

Generated from the .aux files on **17 September 2026**; regenerate before any citation of a printed number, since the counters shift whenever a source is edited. Q91, Q92 and Q147 use a flat theorem

counter; Q149, Q150c1 and Q150c2 number by section.

file	label	number	what it is
Q91	thm:main	7	fermionic normal form
Q92	thm:matrix	5	the two sectors, single t
Q92	thm:main	6	$[R_e, R_f] = -\frac{1+t}{t}(\Phi + (t-1)\Psi)$
Q92	prop:psiwitness	16	the $t^f - t^{f-2}$ witness (two-bead)
Q147	lem:onebead	5	one-bead element, height form
Q147	thm:main	7	height weights force MN
Q149	lem:dict	1.3	word $\leftrightarrow$ composition
Q149	lem:transpose	1.6	transposition = reverse-and-complement
Q149	lem:sectors	4.1	the two sectors, shape form
Q149	lem:onebead	4.2	one-bead element, shape form
Q149	lem:twobead	4.3	two-bead element, shape form
Q149	thm:main	5.2	superseded classification (C2)
Q149	cor:size	5.4	size of the γ -torus
Q149	lem:nonvanish	8.2	false for $\gcd \geq 3$ (C2)
Q150c1	conv:otimes	1.1	concatenation
Q150c1	thm:structure0	1.2	Lyndon–Schützenberger for $\otimes$
Q150c1	thm:main0	1.3	the classification
Q150c1	cor:main0	1.4	corrected nonvanishing
Q150c1	conv:2Bstar	2.1	$(2B^*)$, $(2B^{**})$
Q150c1	prop:1B	3.1	one-bead = $\otimes$ -commutation
Q150c1	prop:2B	3.3	two-bead = $(2B^*)$
Q150c1	thm:gcdle2	6.1	nonvanishing for $\gcd \leq 2$
Q150c2	def:S	2.2	admissible support
Q150c2	prop:supp	2.3	support of a $(2B^*)$ weight is admissible
Q150c2	def:border	4.1	(BS), there called “non-overlapping” (C1)
Q150c2	thm:singleton	4.2	$\not\models_w$ satisfies $(2B^*)$ iff (BS)
Q150c2	thm:singleton2	4.5	$(2B^{**})$ iff antipalindrome
Q150c2	thm:main	5.1	single-shape commuting operators
Q150c2	cor:48	5.2	the smallest new case, $d = 4$
Q150c2	thm:struct	6.1	admissible $\Rightarrow$ inside a cylinder
Q150c2	cor:maximal	6.3	maximal proper = cylinders
Q150c2	cor:quarter	6.4	the dichotomy; prose needs (C3)
Q150c2	rem:sym	6.6	the word-level $(\mathbb{Z}/2)^2$
Q150c2	thm:closure	7.2	generation by the flip rule

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